

NIFTY 100 FINANCIAL INTELLIGENCE PLATFORM

ANALYST USER GUIDE & TECHNICAL MANUAL

1. Executive Summary & Introduction

Welcome to the Nifty 100 Financial Intelligence Platform — an end-to-end data engineering, fundamental analytics, and quantitative intelligence platform built for financial analysts, equity researchers, and portfolio managers. The platform processes 10+ years of historical financial statement data across 92 constituent companies of the Nifty 100 Index.

This comprehensive guide details the operational workflows, Streamlit dashboard screens, stock screening strategies, automated PDF tearsheet generation, REST API endpoints, performance tuning, and troubleshooting procedures.

Platform Module	Primary Component	Key Technologies
ETL Pipeline	Data cleaner & normalizer	Python, Pandas, SQLite3
Financial Ratio Engine	50+ KPI Calculator	NumPy, SciPy
Investment Screener	Preset & custom filter engine	YAML, Pandas Query
Streamlit Dashboard	8 Interactive Screens	Streamlit, Plotly
REST API	16 FastAPI endpoints	FastAPI, Uvicorn, Pydantic
ML Clustering	KMeans 5-cluster model	scikit-learn, StandardScaler
Reporting Engine	Tearsheet & summary PDFs	ReportLab, Matplotlib

2. System Setup & Launch Instructions

Follow these quick-start steps to set up and launch the platform environment locally:

Step 1: Clone Repository & Create Virtual Environment

```
python3 -m venv .venv  
source .venv/bin/activate # On Windows: .venv\Scripts\activate
```

Step 2: Install Dependencies

```
pip install -r requirements.txt
```

Step 3: Build SQLite Database (ETL Load)

```
python src/etl/loader.py
```

Step 4: Compute Ratios & Populated Tables

```
python src/analytics/ratio_engine.py
```

Step 5: Launch Streamlit Analytics Dashboard

```
streamlit run src/dashboard/app.py
```

Step 6: Start REST API Server

```
uvicorn src.api.main:app --port 8000 --reload
```

3. Dashboard Navigation — Screens 1 & 2

Screen 01: Executive Overview (pages/01_home.py)

The Home Screen provides a high-level executive dashboard featuring 6 key summary KPI tiles (Average ROE, Median P/E, Median D/E, Average OPM, FCF Positive Count, Total Universe Market Cap). It includes a Plotly sector distribution donut chart, a Top-5 Quality Leaderboard, and a dynamic FY year selector slider (FY19–FY24).

Screen 02: Company Profile (pages/02_profile.py)

The Company Profile screen offers deep-dive analysis for individual stock tickers. Features include ticker search autocomplete, company header metadata (Logo, Business Description, BSE/NSE links), 6 KPI tiles, 10-Year Revenue & Net Profit bar chart, 10-Year ROE & ROCE trendline chart, and automated Pros & Cons badges.

4. Dashboard Navigation — Screens 3 & 4

Screen 03: Investment Screener (pages/03_screener.py)

The Stock Screener allows multi-parameter filtering across 15+ financial metrics using sidebar range sliders. It includes 6 preset strategy buttons (Buffett Style, Benjamin Graham, Peter Lynch, Quality Compounders, Low Debt, Dividend Growth), a live-updating results table with sortable columns, and an instant CSV download exporter.

Screen 04: Peer Comparison (pages/04_peers.py)

The Peer Comparison module enables head-to-head analysis across 11 industry peer groups (e.g. Private Banks, IT Services, FMCG). It displays an 8-axis Scatterpolar radar chart comparing company performance against peer group average, alongside a side-by-side benchmark comparison table.

5. Dashboard Navigation — Screens 5 & 6

Screen 05: Trend Analysis ([pages/05_trends.py](#))

The Multi-Metric Trend Analysis screen allows analysts to select any company and overlay up to 3 financial metrics over a 10-year historical window. Features YoY percentage change annotations, sparklines, and growth trajectory visualizers.

Screen 06: Sector Analysis ([pages/06_sectors.py](#))

The Sector Analysis screen provides sector rotation and relative positioning insights. Features an interactive Plotly bubble chart (X=Revenue, Y=ROE, Size=Market Cap, Color=Sub-Sector) and sector median financial ratio comparison bar charts across all 11 broad sectors.

6. Dashboard Navigation — Screens 7 & 8

Screen 07: Capital Allocation Map (pages/07_capital.py)

The Capital Allocation Map visualizes how companies generate and deploy cash flow across 8 capital allocation archetypes (Reinvestor, Shareholder Returns, Deleveraging, Cash Accumulator, Growth Funded by Debt, Liquidating Assets, Distress Signal, Pre-Revenue). Features a full-universe Plotly Treemap with click-to-filter drilldowns.

Screen 08: Annual Reports Repository (pages/08_reports.py)

The Annual Reports Repository provides direct access to 1,585 annual report PDF filing URLs sourced from BSE India. Includes ticker filter dropdowns, year selectors, and URL validation badges.

7. Investment Screener Strategy Reference

The platform provides 6 legendary investment strategy presets pre-configured in config/screener_config.yaml:

Preset Name	Core Filter Criteria	Ranking Metric
Buffett Style	ROE > 20%, D/E < 0.5, OPM > 18%, FCF > 0	Composite Score (desc)
Benjamin Graham	P/E < 20, P/B < 2.0, D/E < 1.0, Current Ratio > 1.5	FCF Yield (desc)
Peter Lynch	PAT CAGR 5yr > 18%, Revenue CAGR > 15%, ROE > 18%	PAT CAGR 5yr (desc)
Quality Compounders	ROE > 15%, D/E < 1.0, FCF > 0, Rev CAGR > 10%	Composite Score (desc)
Low Debt Champions	D/E = 0, ROE > 12%, Revenue > 5,000 Cr	ROE (desc)
Dividend Growth	Dividend Yield > 2%, Payout < 80%, FCF > 0	Dividend Yield (desc)

8. REST API Reference & Example Commands

The platform exposes 16 FastAPI REST endpoints running on port 8000 under the /api/v1 prefix:

Example 1: Health Check Endpoint

```
curl -X GET http://localhost:8000/api/v1/health
```

Example 2: Get List of Companies

```
curl -X GET 'http://localhost:8000/api/v1/companies?sector=Information%20Technology'
```

Example 3: Run Stock Screener Filter

```
curl -X GET 'http://localhost:8000/api/v1/screener?min_roe=15&max_de=1.0'
```

Example 4: Download PDF Tearsheet Binary

```
curl -X GET 'http://localhost:8000/api/v1/companies/TCS/tearsheet' --output TCS_tearsheet.pdf
```


9. ML Clustering & Archetype Profiles

KMeans clustering (n_clusters=5, random_state=42) segments all 92 companies based on StandardScaler normalized financial profiles (ROE, D/E, Revenue CAGR 5yr, FCF CAGR 5yr, OPM).

Cluster ID	Archetype Name	Key Financial Characteristics
Cluster 0	High-Quality Compounders	High ROE (>25%), Low D/E (<0.3), Strong FCF Generation
Cluster 1	Defensive Dividend Payers	Stable Cash Flow, High Dividend Yield, Moderate Growth
Cluster 2	Value & Commodity Cyclicals	Cyclical OPM, Moderate Debt, Volatile Earnings
Cluster 3	High-Leverage Utilities/PSUs	High Debt-to-Equity (>1.5), Capital Intensive, Regulated
Cluster 4	Emerging Fast Growers	High Revenue CAGR (>20%), Reinvesting Cash Flow

10. Troubleshooting & FAQ

Issue 1: SQLite Database Connection Errors

Resolution: Ensure db/nifty100.db exists. Re-run 'python src/etl/loader.py' to rebuild SQLite tables.

Issue 2: Port 8000 or 8501 Already in Use

Resolution: Specify custom ports using 'uvicorn src.api.main:app --port 8001' or 'streamlit run src/dashboard/app.py --server.port 8502'.

Issue 3: PDF Generation Overflow in ReportLab

Resolution: Verify ReportLab is version ≥ 4.1 . Table cells auto-wrap text using Paragraph flowables.